

(ii) Change in sales from September to October is given by

$$A - B = \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 5000 & 10,000 & 24,000 \\ 30,000 & 20,000 & 0 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(iii) $2\% \text{ of } B = \frac{2}{100} \times B = 0.02 \times B$

$$= 0.02 \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 5000 & 10,000 & 6000 \\ 20,000 & 10,000 & 10,000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

$$= \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 100 & 200 & 120 \\ 400 & 200 & 200 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

Thus, in October Ramkishan receives ₹ 100, ₹ 200 and ₹ 120 as profit in the sale of each variety of rice, respectively, and Gurcharan Singh receives profit of ₹ 400, ₹ 200 and ₹ 200 in the sale of each variety of rice, respectively.

3.4.5 Multiplication of matrices

Suppose Meera and Nadeem are two friends. Meera wants to buy 2 pens and 5 story books, while Nadeem needs 8 pens and 10 story books. They both go to a shop to enquire about the rates which are quoted as follows:

Pen – ₹ 5 each, story book – ₹ 50 each.

How much money does each need to spend? Clearly, Meera needs ₹ $(5 \times 2 + 50 \times 5)$ that is ₹ 260, while Nadeem needs $(8 \times 5 + 50 \times 10)$ ₹, that is ₹ 540. In terms of matrix representation, we can write the above information as follows:

Requirements Prices per piece (in Rupees) Money needed (in Rupees)

$$\begin{bmatrix} 2 & 5 \\ 8 & 10 \end{bmatrix} \begin{bmatrix} 5 \\ 50 \end{bmatrix} = \begin{bmatrix} 5 \times 2 + 5 \times 50 \\ 8 \times 5 + 10 \times 50 \end{bmatrix} = \begin{bmatrix} 260 \\ 540 \end{bmatrix}$$

Suppose that they enquire about the rates from another shop, quoted as follows:

pen – ₹ 4 each, story book – ₹ 40 each.

Now, the money required by Meera and Nadeem to make purchases will be respectively ₹ $(4 \times 2 + 40 \times 5) = ₹ 208$ and ₹ $(8 \times 4 + 10 \times 40) = ₹ 432$

Again, the above information can be represented as follows:

Requirements Prices per piece (in Rupees) Money needed (in Rupees)

$$\begin{bmatrix} 2 & 5 \\ 8 & 10 \end{bmatrix} \quad \begin{bmatrix} 4 \\ 40 \end{bmatrix} \quad \begin{bmatrix} 4 \times 2 + 40 \times 5 \\ 8 \times 4 + 10 \times 40 \end{bmatrix} = \begin{bmatrix} 208 \\ 432 \end{bmatrix}$$

Now, the information in both the cases can be combined and expressed in terms of matrices as follows:

Requirements Prices per piece (in Rupees) Money needed (in Rupees)

$$\begin{bmatrix} 2 & 5 \\ 8 & 10 \end{bmatrix} \quad \begin{bmatrix} 5 & 4 \\ 50 & 40 \end{bmatrix} \quad \begin{bmatrix} 5 \times 2 + 5 \times 50 & 4 \times 2 + 40 \times 5 \\ 8 \times 5 + 10 \times 50 & 8 \times 4 + 10 \times 40 \end{bmatrix} = \begin{bmatrix} 260 & 208 \\ 540 & 432 \end{bmatrix}$$

The above is an example of multiplication of matrices. We observe that, for multiplication of two matrices A and B, the number of columns in A should be equal to the number of rows in B. Furthermore for getting the elements of the product matrix, we take rows of A and columns of B, multiply them element-wise and take the sum. Formally, we define multiplication of matrices as follows:

The *product* of two matrices A and B is *defined* if the number of columns of A is equal to the number of rows of B. Let $A = [a_{ij}]$ be an $m \times n$ matrix and $B = [b_{jk}]$ be an $n \times p$ matrix. Then the product of the matrices A and B is the matrix C of order $m \times p$. To get the $(i, k)^{\text{th}}$ element c_{ik} of the matrix C, we take the i^{th} row of A and k^{th} column of B, multiply them elementwise and take the sum of all these products. In other words, if $A = [a_{ij}]_{m \times n}$, $B = [b_{jk}]_{n \times p}$, then the i^{th} row of A is $[a_{i1} \ a_{i2} \ \dots \ a_{in}]$ and the k^{th} column of

$$B \text{ is } \begin{bmatrix} b_{1k} \\ b_{2k} \\ \vdots \\ b_{nk} \end{bmatrix}, \text{ then } c_{ik} = a_{i1} b_{1k} + a_{i2} b_{2k} + a_{i3} b_{3k} + \dots + a_{in} b_{nk} = \sum_{j=1}^n a_{ij} b_{jk}.$$

The matrix $C = [c_{ik}]_{m \times p}$ is the product of A and B.

$$\text{For example, if } C = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix} \text{ and } D = \begin{bmatrix} 2 & 7 \\ -1 & 1 \\ 5 & -4 \end{bmatrix}, \text{ then the product CD is defined}$$